

Air Quality and Health in the United States

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Overview

This project explores patterns of air quality across the United States primarily using the Air Quality Index (AQI) data maintained by the EPA(The United States Environmental Protection Agency). We first conducted Exploratory Data Analysis to explore basic patterns in the data with python and Tableau, and then moved on to our Main questions.

Main Questions and Methodology

Q1: How does air quality vary by Urbanization Levels?

- We will use an additional dataset(Urbanization Score) alongside the AQI dataset to analyze this question through visualizations.

Q2: Can We Identify Areas Where Poor Air Quality May Have the Most Severe Impact?

- We will use an additional dataset(Social Vulnerability Index) alongside the AQI dataset and introduce a new metric to analyze this.

Exploratory Data Analysis

Description of dataset

Dataset: [AQI.csv](#) (The main dataset provided by the datathon)

The EPA Air Quality Index (AQI) by County dataset is maintained by the EPA(The United States Environmental Protection Agency), a federal agency responsible for protecting human health and the environment. It contains annual air quality metrics for every U.S. county, including counts of good, moderate, unhealthy, and hazardous days, as well as pollutant-specific exposure (CO, NO₂, Ozone, PM2.5, PM10). All of this data comes from EPA's Air Quality System (AQS), where data collection agencies report their data to EPA via this system and calculate several types of aggregate data for EPA internal use. This is the 2025 Dataset.

(A brief overview of the dataframe)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('AQI.csv')
df.head()
```

	State	County	Year	Days with AQI	Good Days	Moderate Days	Unhealthy for Sensitive Groups Days	Unhealthy Days	Very Unhealthy Days	Hazardous Days	Max AQI	90th Percentile AQI	Median AQI	Days CO	Days NO2	Days Ozone	Days PM2.5	Days PM10
0	Alabama	Baldwin	2025	241	174	67	0	0	0	0	87	56	42	0	0	91	150	0
1	Alabama	Clay	2025	239	204	34	1	0	0	0	133	52	32	0	0	0	239	0
2	Alabama	DeKalb	2025	243	191	52	0	0	0	0	93	55	42	0	0	156	87	0
3	Alabama	Elmore	2025	177	172	5	0	0	0	0	64	46	32	0	0	177	0	0
4	Alabama	Etowah	2025	241	153	88	0	0	0	0	87	58	45	0	0	72	169	0

↑ Description of columns:

State	U.S. State
County	County name
Year	Year of measurement
Days with AQI	Total days monitored
Good Days (AQI 0-50) Moderate Days (AQI 51-100) Unhealthy for Sensitive Groups Days (AQI 101-150) Unhealthy Days (AQI 151-200) Very Unhealthy Days (AQI 201-300) Hazardous Days (AQI 301+)	Exposure metrics
Max AQI / 90th Percentile AQI / Median AQI	Severity measures
Days CO / Days NO2 / Days Ozone / Days PM2.5 / Days PM10	Pollutant-specific exposure

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 978 entries, 0 to 977
Data columns (total 18 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   State                                     978 non-null    object
1   County                                   978 non-null    object
2   Year                                     978 non-null    int64
3   Days with AQI                           978 non-null    int64
4   Good Days                               978 non-null    int64
5   Moderate Days                           978 non-null    int64
6   Unhealthy for Sensitive Groups Days      978 non-null    int64
7   Unhealthy Days                           978 non-null    int64
8   Very Unhealthy Days                     978 non-null    int64
9   Hazardous Days                           978 non-null    int64
10  Max AQI                                 978 non-null    int64
11  90th Percentile AQI                     978 non-null    int64
12  Median AQI                              978 non-null    int64
13  Days CO                                  978 non-null    int64
14  Days NO2                                 978 non-null    int64
15  Days Ozone                               978 non-null    int64
16  Days PM2.5                               978 non-null    int64
17  Days PM10                                978 non-null    int64
dtypes: int64(16), object(2)
memory usage: 137.7+ KB
```

↓

The dataset includes 978 rows representing counties across the 50 states. However, it does not cover all U.S. counties, which may limit the completeness of county-level comparisons.

Preprocessing

First, non-U.S. states including Mexico and U.S. territories, were excluded to ensure consistency and convenience with later work. ↓

```
territories = [
    "Virgin Islands",
    "Country Of Mexico",
    "Puerto Rico",
    "Guam",
    "American Samoa",
    "Northern Mariana Islands"
]

df = df[~df["State"].isin(territories)]
```

We want to find states/counties whose AQIs are high, but simply aggregating the number of high AQI days would be misleading since each county has a different number of days with AQI measurements. Therefore, we defined days with AQI at "Unhealthy for Sensitive Groups" level or higher as "unhealthy days" and created a new column containing the percentage relative to total measurement days. ↓

```
unhealthy_cols = ['Unhealthy for Sensitive Groups Days', 'Unhealthy Days', 'Very Unhealthy Days', 'Hazardous Days']
df['Pct_Unhealthy_Days'] = df[unhealthy_cols].sum(axis=1) * 100 / df['Days with AQI']
df
```

Year	Days with AQI	Good Days	Moderate Days	Unhealthy for Sensitive Groups Days	Unhealthy Days	Very Unhealthy Days	Hazardous Days	Max AQI	90th Percentile AQI	Median AQI	Days CO	Days NO2	Days Ozone	Days PM2.5	Days PM10	Pct_Unhealthy_Days
2025	241	174	67	0	0	0	0	87	56	42	0	0	91	150	0	0.000000
2025	239	204	34	1	0	0	0	133	52	32	0	0	0	239	0	0.418410
2025	243	191	52	0	0	0	0	93	55	42	0	0	156	87	0	0.000000
2025	177	172	5	0	0	0	0	64	46	32	0	0	177	0	0	0.000000
2025	241	153	88	0	0	0	0	87	58	45	0	0	72	169	0	0.000000
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2025	242	165	76	1	0	0	0	101	67	47	0	0	240	2	0	0.413223
2025	212	149	59	4	0	0	0	115	64	45	0	0	164	30	18	1.886792
2025	226	179	47	0	0	0	0	84	58	45	2	0	221	3	0	0.000000
2025	87	87	0	0	0	0	0	50	21	10	0	0	0	0	87	0.000000
2025	202	198	4	0	0	0	0	59	23	10	0	0	0	202	0	0.000000

And to compare air quality conditions across states, we aggregated county-level air quality data to the state level. ↓

```
state_summary = (
    df.groupby("State").agg(
        avg_pct_unhealthy = ("Pct_Unhealthy_Days", "mean"),
        avg_max_aqi = ("Max AQI", "mean"),
        avg_90th_aqi = ("90th Percentile AQI", "mean"),
        avg_median_aqi = ("Median AQI", "mean"),

        counties = ("County", "count")
    )
    .sort_values("avg_pct_unhealthy", ascending=False)
)
```

Some Basic Visualizations

```
state_summary = (
    df.groupby("State").agg(
        avg_pct_unhealthy = ("Pct_Unhealthy_Days", "mean"),
        avg_max_aqi = ("Max AQI", "mean"),
        avg_90th_aqi = ("90th Percentile AQI", "mean"),
        avg_median_aqi = ("Median AQI", "mean"),

        counties = ("County", "count")
    )
    .sort_values("avg_pct_unhealthy", ascending=False)
)

sns.set_theme(style="whitegrid")

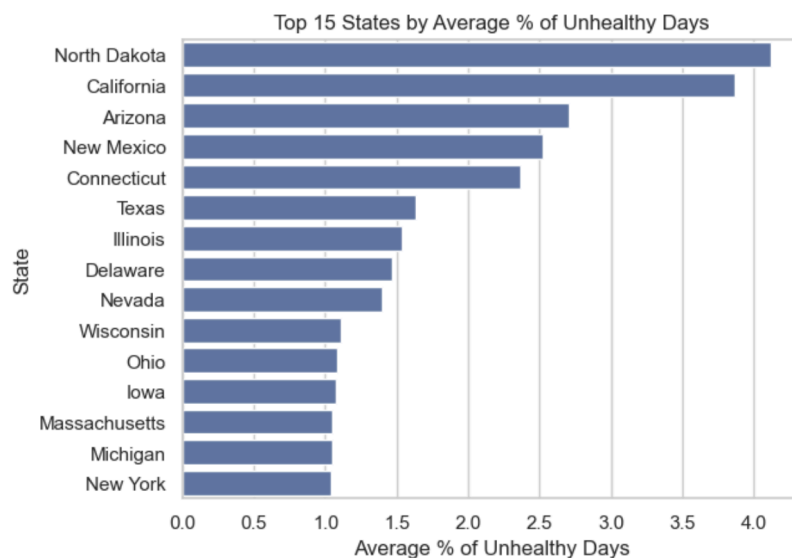
top_n_states = 15

state_plot = (
    state_summary
    .reset_index()
    .sort_values("avg_pct_unhealthy", ascending=False)
    .head(top_n_states)
)

sns.barplot(
    data = state_plot,
    x = "avg_pct_unhealthy",
    y = "State"
)

plt.xlabel("Average % of Unhealthy Days")
plt.ylabel("State")
plt.title(f"Top {top_n_states} States by Average % of Unhealthy Days")
plt.show()
```

We first created a bar chart using the `state_summary` we created to visualize the states with the highest average percentage of unhealthy days.



-> North Dakota had the highest average percentage of unhealthy days in 2025, followed by California, Arizona, and New Mexico.

```

top_n_counties = 20

df["County_State"] = df["County"] + ", " + df["State"]

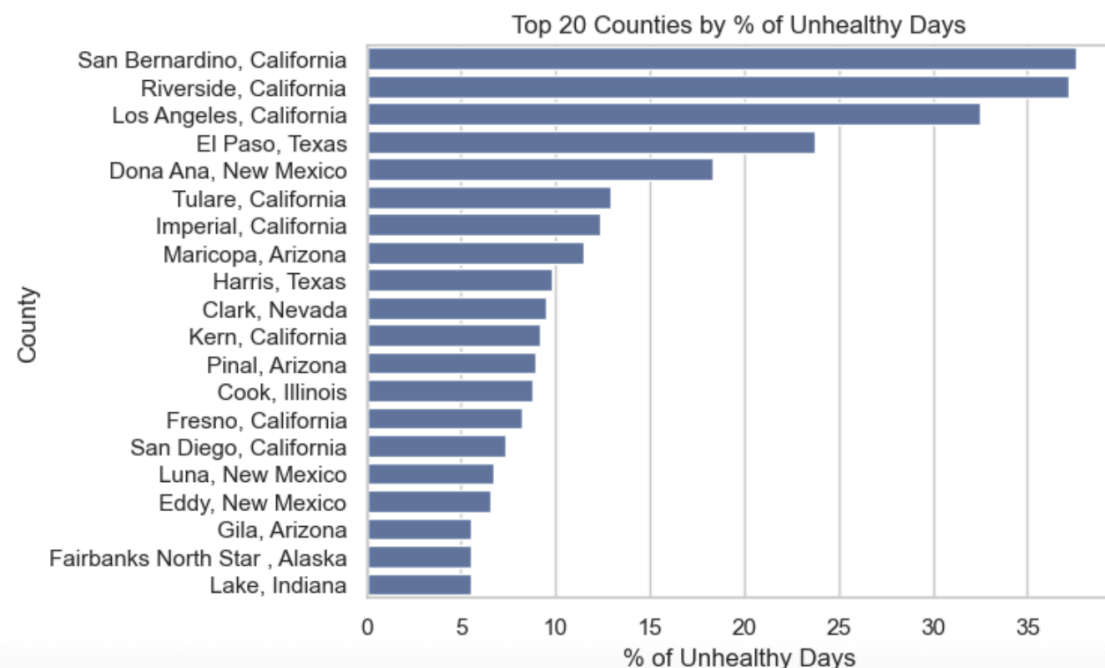
county_plot = (
    df.sort_values("Pct_Unhealthy_Days", ascending=False)
    .head(top_n_counties)
)

sns.barplot(
    data = county_plot,
    x = "Pct_Unhealthy_Days",
    y = "County_State"
)

plt.xlabel("% of Unhealthy Days")
plt.ylabel("County")
plt.title(f"Top {top_n_counties} Counties by % of Unhealthy Days")
plt.show()

```

We then examined county-level patterns by visualizing the top 20 counties with the highest percentage of unhealthy air quality days.



-> The top 20 counties include multiple counties from California such as the San Bernardino County and Riverside County, as well as counties from Texas, Arizona, New Mexico, Nevada, Alaska, and Indiana. California, Arizona, and New Mexico are frequent guest states in this graph.

To get a better geographical sense of the U.S.'s air quality situations, we created an interactive Tableau dashboard map, accessible via the link below:

https://public.tableau.com/shared/XGGXP5N4T?:display_count=n&origin=viz_share_link

Q1: How does air quality vary by Urbanization Levels?

Overview

Cities typically experience higher traffic density and population concentration, which may contribute to increased emissions from transportation, higher energy demand, and other pollution generating activities. We will explore whether the level of urbanization plays a role in the air quality of that area.

Additional Dataset: [urbanization.csv](#)

- In addition to [AQI.csv](#), we will use [urbanization.csv](#), which contains the urbanization level of each U.S. county.
- The National Center for Health Statistics (NCHS) has developed a six-level urban-rural classification scheme for U.S. counties:
 - 1 - Large central metro (1m+ population & other qualifications)
 - 2 - Large fringe metro (1m+ population but doesn't qualify for Large central metro)
 - 3 - Medium metro (250,000 to 999,999 population)
 - 4 - Small metro (50,000 to 249,999 population)
 - 5 - Micropolitan (counties in micropolitan statistical areas)
 - 6 - Noncore (did not qualify as micropolitan)
- It classifies 3,144 U.S. counties, including county equivalents such as parishes in Louisiana, boroughs in Alaska, and planning regions in Connecticut, but not including county equivalents in U.S. territories.
- Using this dataset, we will look at the correlation of the classification level of U.S. counties with the air quality index.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
urbanization = pd.read_csv('urbanization.csv')
urbanization.head()
```

	FullGeoName	Location	State	County_name	2023 Code
0	AK, Aleutians East Borough	2013	AK	Aleutians East Borough	6 - Noncore
1	AK, Aleutians West Ca	2016	AK	Aleutians West Census Area	6 - Noncore
2	AK, Anchorage Muny	2020	AK	Anchorage Municipality	3 - Medium metro
3	AK, Bethel Ca	2050	AK	Bethel Census Area	6 - Noncore
4	AK, Bristol Bay Borough	2060	AK	Bristol Bay Borough	6 - Noncore

```
urbanization.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3144 entries, 0 to 3143
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   FullGeoName  3144 non-null   object
1   Location     3144 non-null   int64
2   State        3144 non-null   object
3   County_name  3144 non-null   object
4   2023 Code    3144 non-null   object
dtypes: int64(1), object(4)
memory usage: 122.9+ KB
```

Preprocessing

To link air quality data with urbanization scores, we had to merge AQI.csv with urbanization.csv. Since the urbanization dataset contains a much larger number of counties, we used a left join to add urbanization scores only for counties in the AQI dataset.

The process was really complicated to do, but it looks like below. The bottom idea was to use both state and county names as merge keys(since county names can overlap with state names, like Washington) and standardize state/county names in both datasets before performing a merge.

```
# change AQI.csv's state names to abbreviations so that merge is available
us_state_to_abbreviation = {
    "Alabama": "AL", "Alaska": "AK", "Arizona": "AZ", "Arkansas": "AR", "California": "CA",
    "Colorado": "CO", "Connecticut": "CT", "Delaware": "DE", "Florida": "FL", "Georgia": "GA",
    "Hawaii": "HI", "Idaho": "ID", "Illinois": "IL", "Indiana": "IN", "Iowa": "IA",
    "Kansas": "KS", "Kentucky": "KY", "Louisiana": "LA", "Maine": "ME", "Maryland": "MD",
    "Massachusetts": "MA", "Michigan": "MI", "Minnesota": "MN", "Mississippi": "MS", "Missouri": "MO",
    "Montana": "MT", "Nebraska": "NE", "Nevada": "NV", "New Hampshire": "NH", "New Jersey": "NJ",
    "New Mexico": "NM", "New York": "NY", "North Carolina": "NC", "North Dakota": "ND", "Ohio": "OH",
    "Oklahoma": "OK", "Oregon": "OR", "Pennsylvania": "PA", "Rhode Island": "RI", "South Carolina": "SC",
    "South Dakota": "SD", "Tennessee": "TN", "Texas": "TX", "Utah": "UT", "Vermont": "VT",
    "Virginia": "VA", "Washington": "WA", "West Virginia": "WV", "Wisconsin": "WI", "Wyoming": "WY",
    "District of Columbia": "DC", "District Of Columbia": "DC",
}

df_aqi['State_Abbreviated'] = df_aqi['State'].str.strip().map(us_state_to_abbreviation)

# clean AQI.csv's county names so that merge is available
df_aqi['Normalized_County'] = df_aqi['County'].str.strip()\
    .str.replace(r' \((City)\)', ' City', regex = False)\
    .str.replace(r'^Saint ', 'St. ', regex = False)\
    .str.replace(r'^Sainte ', 'Ste. ', regex = False)

# clean urbanization.csv's county names so that merge is available
df_urb['County_Cleaned'] = df_urb['County_name'].str.strip()\
    .str.replace(' City and Borough', '', regex = False)\
    .str.replace(' Municipality', '', regex = False)\
    .str.replace(' Census Area', '', regex = False)\
    .str.replace(' Borough', '', regex = False)\
    .str.replace(' Parish', '', regex = False)\
    .str.replace(' County', '', regex = False)\
    .str.replace(' city', ' City', regex = False)

merged_df = pd.merge(df_aqi, df_urb[['State', 'County_Cleaned', '2023 Code']],
    left_on = ['State_Abbreviated', 'Normalized_County'],
    right_on = ['State', 'County_Cleaned'],
    how = 'left')

merged_df = merged_df.rename(columns={'2023 Code': 'Urbanization Score'})
columns_to_drop = ['State_Abbreviated', 'Normalized_County', 'State_y', 'County_Cleaned']
merged_df = merged_df.drop(columns=columns_to_drop)
merged_df['Urbanization Score'] = pd.to_numeric(merged_df['Urbanization Score'].str[0])

merged_df.describe()
```

Very Healthy Days	Hazardous Days	Max AQI	90th Percentile AQI	Median AQI	Days CO	Days NO2	Days Ozone	Days PM2.5	Days PM10	Pct_Unhealthy_Days	Urbanization Score
.000000	968.000000	968.000000	968.000000	968.000000	968.000000	968.000000	968.000000	968.000000	968.000000	968.000000	953.000000
.040289	0.043388	113.236570	57.712810	38.653926	0.609504	1.798554	116.029959	92.908058	6.174587	0.849542	3.729276
.323867	0.534418	93.517148	14.497447	10.492676	5.734418	13.423024	86.186806	79.707939	29.173118	2.563590	1.529182
.000000	0.000000	8.000000	8.000000	3.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000
.000000	0.000000	80.000000	51.000000	36.000000	0.000000	0.000000	43.000000	6.000000	0.000000	0.000000	3.000000
.000000	0.000000	99.000000	58.000000	40.500000	0.000000	0.000000	113.000000	88.000000	0.000000	0.000000	4.000000
.000000	0.000000	126.250000	64.000000	44.000000	0.000000	0.000000	187.000000	156.000000	0.000000	0.894868	5.000000
.000000	12.000000	2122.000000	164.000000	90.000000	94.000000	241.000000	304.000000	304.000000	288.000000	37.591241	6.000000

Visualization

We used two visualization types here. First, to understand the overall distribution shape (density, skewness) of the data, we used a violin plot. Second, we used a pointplot to intuitively show the trend of change in the mean.

```
plt.figure(figsize=(12, 6))
sns.set_theme(style = 'whitegrid')

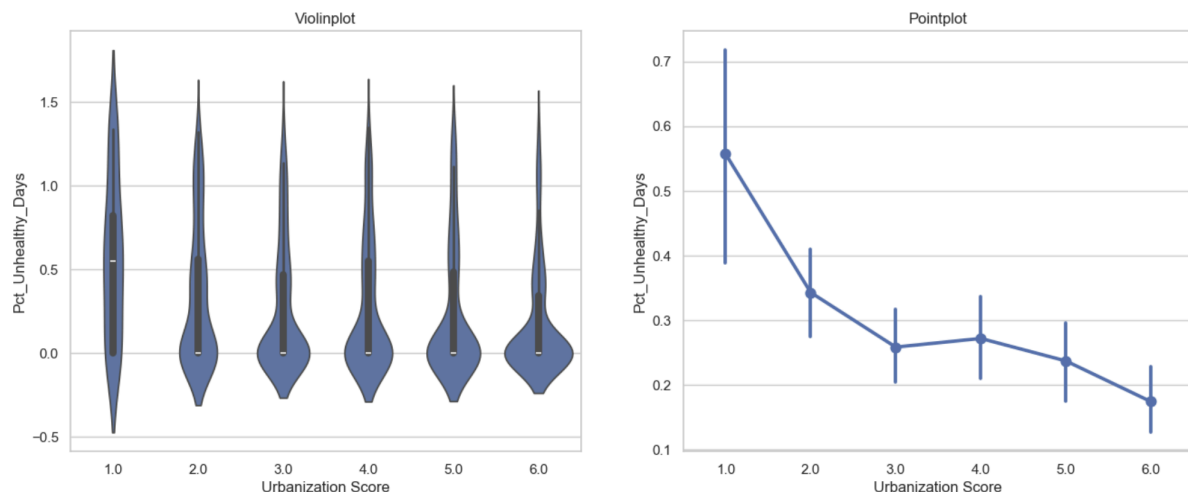
q1 = merged_df['Pct_Unhealthy_Days'].quantile(0.25)
q3 = merged_df['Pct_Unhealthy_Days'].quantile(0.75)
iqr = q3 - q1
lower_limit = q1 - 1.5 * iqr
upper_limit = q3 + 1.5 * iqr

merged_df = merged_df[(merged_df['Pct_Unhealthy_Days'] >= lower_limit) & (merged_df['Pct_Unhealthy_Days'] <= upper_limit)]

fig, (ax0, ax1) = plt.subplots(1, 2, figsize=(16, 6))

sns.violinplot(x = 'Urbanization Score', y = 'Pct_Unhealthy_Days', data = merged_df, ax = ax0)
ax0.set_title('Violinplot')

sns.pointplot(x = 'Urbanization Score', y = 'Pct_Unhealthy_Days', data = merged_df, ax = ax1)
ax1.set_title('Pointplot')
```



Q1 Conclusion

We can immediately see that areas with a score of 1 (Large central metro) clearly have the highest median value. Also, the point plot shows that the mean percentage of days defined as 'unhealthy' goes down as the score gets higher (=moving towards the countryside). This confirms that big cities are much more likely to have bad air days compared to rural areas.

We also want to point out variability. For the more urban areas (Scores 1-2), the interquartile range is generally taller. This may suggest there is high variability in these areas; while some large cities are managing air quality well, others could be struggling. Also, for relatively rural areas (Scores 3-6), the interquartile range tends to be shorter and compressed. This may imply that air quality is consistently good across most rural counties with less variation.

Q2. Can We Identify Areas Where Poor Air Quality May Have the Most Severe Impact?

Overview

Some people might be more susceptible to bad air quality. While high pollution levels are a risk to the general public, the actual health impact is dictated by a community's biological sensitivity and its ability to overcome diseases. In this analysis, we will quantify that biological sensitivity using the Social Vulnerability Index dataset, and find areas/regions that are more vulnerable to poor air quality.

Additional Dataset:

- In addition to [AQI.csv](#), we will use [SVI.csv](#), which contains the social vulnerability index data of U.S. counties, up to date until 2022.
- The *Centers for Disease Control and Prevention and Agency for Toxic Substances and Disease Registry* Social Vulnerability Index (CDC/ATSDR SVI or SVI), is a place-based database designed to identify and quantify communities experiencing social vulnerability. The Geospatial Research, Analysis & Services Program maintains the SVI to help public health officials and local planners better prepare for and respond to health related events.

```
df_svi = pd.read_csv('SVI.csv')
df_svi.head()
```

	ST	STATE	ST_ABBR	STCNTY	COUNTY	FIPS	LOCATION	AREA_SQMI	E_TOTPOP	M_TOTPOP	...	EP_ASIAN	MP_ASIAN	EP_AIAN	MP_AIAN	EP_NHPI
0	1	Alabama	AL	1001	Autauga County	1001	Autauga County, Alabama	594.454786	58761	0	...	1.1	0.4	0.1	0.1	0.0
1	1	Alabama	AL	1003	Baldwin County	1003	Baldwin County, Alabama	1589.861817	233420	0	...	0.9	0.1	0.2	0.1	0.0
2	1	Alabama	AL	1005	Barbour County	1005	Barbour County, Alabama	885.007619	24877	0	...	0.5	0.1	0.3	0.1	0.0
3	1	Alabama	AL	1007	Bibb County	1007	Bibb County, Alabama	622.469286	22251	0	...	0.3	0.4	0.1	0.1	0.0
4	1	Alabama	AL	1009	Blount County	1009	Blount County, Alabama	644.890376	59077	0	...	0.2	0.2	0.1	0.1	0.2

- This dataset contains a large number of columns(158), and they are all different indicators of social vulnerability. This includes measures related to socioeconomic status, household characteristics etc. It was designed for responding to physical disasters such as floods or earthquakes and therefore combines more than 16 variables, often emphasizing factors such as mobile home residency and vehicle ownership.
- From these indicators, we will create a new metric to represent the population's biological/economical vulnerability to poor air quality. Using this metric, we will examine how vulnerable U.S. counties are to poor air quality, and the possible threat caused to them.

Metric Construction: Air Health Vulnerability Index (AHVI)

For air quality analysis, factors that are related to respiratory health are more relevant. This metric is calculated as the mean of 5 specific percentile ranks from the SVI dataset, focusing exclusively on health related features.

EPL_AGE65	(Elderly)	High risk of respiratory failure
EPL_AGE17	(Children)	Developing lungs are more susceptible to damage
EPL_DISABL	(Disability)	May have pre-existing conditions
EPL_POV150	(Poverty)	May lack resources like high-quality masks or the ability to stay indoors
EPL_UNINSUR	(Uninsured)	Inability to access medical care

$$AHVI = \frac{EPL_AGE65 + EPL_AGE17 + EPL_DISABL + EPL_POV150 + EPL_UNINSUR}{5}$$

```
ahvi_columns = ['EPL_AGE65', 'EPL_AGE17', 'EPL_POV150', 'EPL_UNINSUR', 'EPL_DISABL']  
df_svi['AHVI'] = df_svi[ahvi_columns].mean(axis=1)
```

-> Creating AHVI column to the SVI dataframe

```
correlation = df_svi['AHVI'].corr(df_svi['RPL_THEMES'])  
print(correlation)  
  
0.6249044974924078
```

-> The correlation between AHVI and the general SVI (RPL_THEMES) is 0.62. (Using the Pearson Correlation Coefficient) This correlation means a county could be considered generally safe from disasters but vulnerable to air pollution specifically.

```
cedar = df_svi[(df_svi['COUNTY'] == 'Cedar County') & (df_svi['STATE'] == 'Missouri')].iloc[0]  
  
ahvi_rank = (df_svi['AHVI'] < cedar['AHVI']).mean() * 100  
svi_rank = (df_svi['RPL_THEMES'] < cedar['RPL_THEMES']).mean() * 100  
  
print(ahvi_rank, svi_rank)  
  
99.87277353689568 66.41221374045801
```

-> For example, Cedar County, MO has a general SVI score in the 66th percentile (moderate), but its AHVI is 0.99, which is extremely high. This suggests that while it may not be seen as a vulnerable area in general terms, its population structure (likely elderly or low-income) makes air quality events much more dangerous there.

Ranking Counties based on AHVI

```
top_vulnerable = df_svi[['STATE', 'COUNTY', 'AHVI', 'RPL_THEMES']].sort_values(by='AHVI', ascending=False)

print("Top 10 U.S. Counties Most Vulnerable to Air Quality Health Impacts:")
print(top_vulnerable.head(20).to_string(index=False))
```

Top 10 U.S. Counties Most Vulnerable to Air Quality Health Impacts:

STATE	COUNTY	AHVI	RPL_THEMES
Texas	Kenedy County	0.93458	0.6678
Texas	Real County	0.92764	0.7830
Oklahoma	Coal County	0.86018	0.8358
Missouri	Cedar County	0.85692	0.6643
Missouri	Oregon County	0.85086	0.6589
Oklahoma	Jefferson County	0.84460	0.8056
Texas	Cottle County	0.84110	0.6732
Texas	Presidio County	0.83766	0.9933
Oklahoma	Choctaw County	0.83656	0.9485
Alabama	Greene County	0.82836	0.9157
Nebraska	Garden County	0.82604	0.4200
Oklahoma	Pushmataha County	0.82004	0.8896
Mississippi	Lawrence County	0.81500	0.8505
Mississippi	Humphreys County	0.81472	0.9787
Texas	San Augustine County	0.81462	0.9885
Missouri	Morgan County	0.80930	0.8657
Georgia	Treutlen County	0.80658	0.9389
Kansas	Chautauqua County	0.80412	0.5781
Oklahoma	Seminole County	0.80298	0.9523
Oklahoma	Haskell County	0.80242	0.8387

Rural counties in states like Texas, Oklahoma, and Missouri are displayed to be most vulnerable to air quality impacts, rather than major urban centers. AHVI suggests that these areas may have the highest concentrations of biologically sensitive groups such as the elderly / disabled / the poor / those with lack of health insurance.

According to the [Rural Health Information Hub](#), rural areas have concentrations of older adults (20%+ age 65), disabled residents (15% higher prevalence), and uninsured populations (3x urban rates).

Preprocessing

Just like the urbanization analysis, we had to merge AQI.csv with SVI.csv. The SVI dataset also contains a much larger number of counties, so we used a left join again.

However, this one is much simpler since SVI.csv includes full state names instead of only having them as abbreviations, and we already have a merged dataframe with cleaned county names. We just have to clean the names of SVI.csv's state and county names.

```
df_svi['State_Cleaned'] = df_svi['STATE'].str.strip()

df_svi['County_Cleaned'] = df_svi['COUNTY'].str.strip()\
    .str.replace(' City and Borough', '', regex=False)\
    .str.replace(' Municipality', '', regex=False)\
    .str.replace(' Census Area', '', regex=False)\
    .str.replace(' Borough', '', regex=False)\
    .str.replace(' Parish', '', regex=False)\
    .str.replace(' County', '', regex=False)\
    .str.replace(' city', ' City', regex=False)

merged_df1 = pd.merge(merged_df, df_svi[['State_Cleaned', 'County_Cleaned', 'AHVI']],
    left_on=['State_x', 'County'],
    right_on=['State_Cleaned', 'County_Cleaned'],
    how='left')

merged_df1 = merged_df1.drop(columns=['State_Cleaned', 'County_Cleaned'])
merged_df1
```

Moderate Days	Unhealthy for Sensitive Groups Days	Unhealthy Days	Very Unhealthy Days	Hazardous Days	...	90th Percentile AQI	Median AQI	Days CO	Days NO2	Days Ozone	Days PM2.5	Days PM10	Pct_Unhealthy_Days	Urbanization Score	AHVI
67	0	0	0	0	...	56	42	0	0	91	150	0	0.000000	4.0	0.44582
34	1	0	0	0	...	52	32	0	0	0	239	0	0.418410	6.0	0.59446
52	0	0	0	0	...	55	42	0	0	156	87	0	0.000000	5.0	0.65204
5	0	0	0	0	...	46	32	0	0	177	0	0	0.000000	3.0	0.37110
88	0	0	0	0	...	58	45	0	0	72	169	0	0.000000	4.0	0.59600
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
76	1	0	0	0	...	67	47	0	0	240	2	0	0.413223	6.0	0.49100
59	4	0	0	0	...	64	45	0	0	164	30	18	1.886792	5.0	0.49944
47	0	0	0	0	...	58	45	2	0	221	3	0	0.000000	5.0	0.22180
0	0	0	0	0	...	21	10	0	0	0	0	87	0.000000	5.0	0.50270
4	0	0	0	0	...	23	10	0	0	0	202	0	0.000000	6.0	0.52154

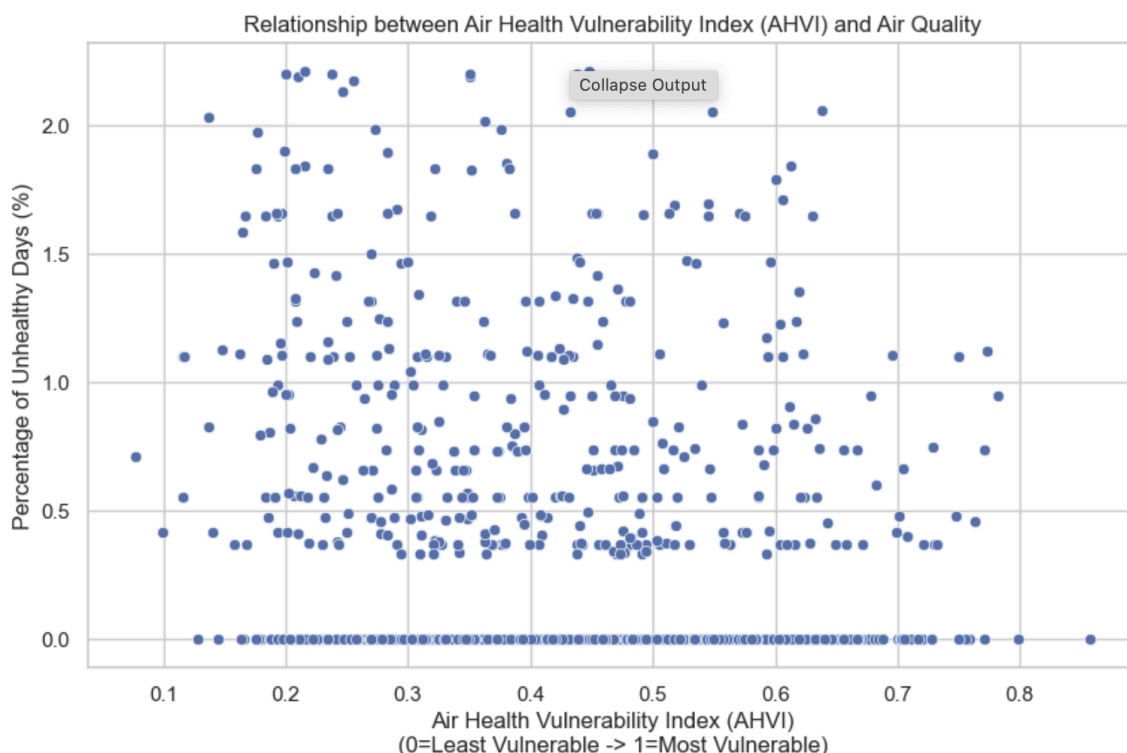
Visualization

```
plt.figure(figsize = (10, 6))
sns.set_theme(style = "whitegrid")

sns.scatterplot(x = 'AHVI', y = 'Pct_Unhealthy_Days', data = merged_df1)

plt.title('Relationship between Air Health Vulnerability Index (AHVI) and Air Quality')
plt.xlabel('Air Health Vulnerability Index (AHVI)\n(0=Least Vulnerable -> 1=Most Vulnerable)')
plt.ylabel('Percentage of Unhealthy Days (%)')
```

Since both axes represent continuous data unlike the discrete urbanization score, boxplot-type visualizations such as the violin plot are not appropriate. We used a scatterplot instead to visualize relationships between two continuous variables. Areas with high AHVIs as well as high Percentage of Unhealthy days will be the most vulnerable.



However, the scatter plot does not show any clear points in the upper-right corner, which would indicate the most dangerous counties. This pattern may be plausible since urban areas with poorer air quality often have lower AHVI scores due to younger population structures and better access to healthcare, while rural areas with higher AHVI scores tend to experience cleaner air. As a result, cases in which both risks overlap may be relatively rare.

Nevertheless, it is difficult to determine which counties face the greatest overall risk based on this visualization alone.

Calculating Risk Scores

So we introduced a new metric called **risk score**, which is $\text{Pct_Unhealthy_Days} \times \text{AHVI}$. It is a simple metric calculated by multiplying the percentage of unhealthy air quality days by the AHVI. It identifies high-risk areas where air pollution exposure intersects with a population's biological / socioeconomic inability.

```
merged_df1['Risk_Score'] = merged_df1['Pct_Unhealthy_Days'] * merged_df1['AHVI']

dangerous_areas = merged_df1[
    ['State_x',
     'County',
     'Risk_Score',
     'Pct_Unhealthy_Days',
     'AHVI']
].sort_values(by='Risk_Score', ascending=False)

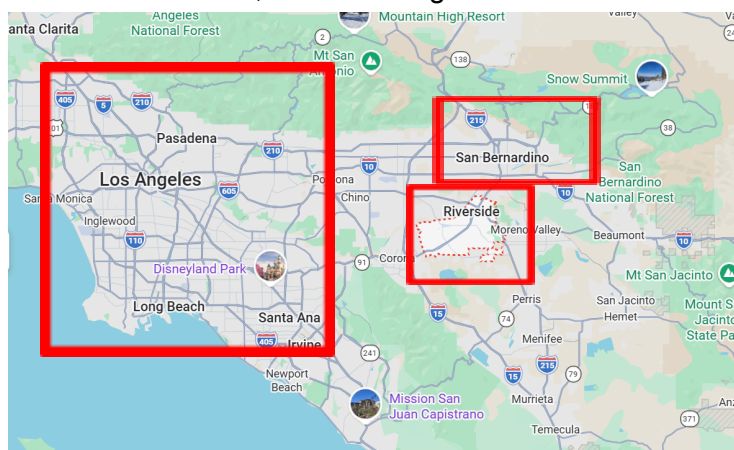
print(dangerous_areas.head(10).to_string(index = False))
```

After calculating, we printed out the top 10 counties in risk score.

State_x	County	Risk_Score	Pct_Unhealthy_Days	AHVI
California	San Bernardino	15.795088	37.591241	0.42018
Texas	El Paso	14.743410	23.770492	0.62024
California	Riverside	14.177255	37.226277	0.38084
California	Los Angeles	10.351285	32.481752	0.31868
California	Imperial	6.502932	12.328767	0.52746
California	Tulare	6.303103	12.931034	0.48744
New Mexico	Luna	5.323567	6.716418	0.79262
Texas	Harris	5.096693	9.854015	0.51722
Arizona	Pinal	4.692179	8.955224	0.52396
Arizona	Maricopa	4.509243	11.513158	0.39166

-> We can see that Southern areas prevail, with California dominating the top spots and Texas / Arizona following. This may reflect higher transportation-related emissions and climatic conditions that exacerbate air pollution in southern states.

While it's interesting that California dominates the top ranks, we found it more intriguing that if you put all the ranked Californian counties in a map, they are all near each other in Southern California, near Los Angeles.



We wondered why this happens, and why this region exhibits such high risk scores.

Probable Cause for Southern California's high risk score

One possible explanation may be the commonness of diesel-power & the region's socioeconomic realities.

First, this area is known as a logistics hub for LA/Long Beach ports, and the prevalence of diesel vehicles whose emissions later turn into ground-level ozone might be the root cause of bad air quality.

```
result = merged_df1.loc[
    (merged_df1['State_x'] == 'California') & (merged_df1['County'].isin(['Los Angeles', 'San Bernardino', 'Riverside'])),
    ['County', 'Days CO', 'Days NO2', 'Days Ozone', 'Days PM2.5', 'Days PM10']
].reset_index(drop=True)
```

```
print(result)
```

	County	Days CO	Days NO2	Days Ozone	Days PM2.5	Days PM10
0	Los Angeles	1	3	172	87	11
1	Riverside	1	0	184	56	33
2	San Bernardino	1	1	201	62	9

↑ And when we checked their pollutant data, we identified they all have especially high Ozone values, all well over the median (113) and near, even above the 75% percentile(187). According to the American Lung Association's "State of the Air 2025", Los Angeles - Long Beach CA ranks first among the 25 cities most polluted by ozone. [ALA 2025](#)

Second, their socioeconomic factors are likely to be playing a big part in their SVI(Social Vulnerability Index)s. According to UC Riverside's study, these counties exhibit high SVI scores due to realities like low-wage warehouse employment, certain ethnic compositions, and high poverty rates. [UCR 2025](#)

While the true causes are likely more complex, this offers a plausible explanation for the high overall risk scores.

Q2 Conclusion

Rural counties in states like Texas, Oklahoma, and Missouri have the highest AHVI scores rather than major urban centers, displaying to be most vulnerable to air quality impacts.

The risk score analysis shows Inland Southern California (San Bernardino and Riverside counties) and Los Angeles as areas facing a double burden, ranking both high in ozone pollution as well as having high AHVIs.

From a healthcare policy perspective, responses should vary by region, with a focus on healthcare access in high-AHVI rural areas and emission reduction in high-risk score regions.

Overall Conclusions

Air quality challenges in the U.S. are intertwined with numerous factors, including urbanization and social vulnerability. There is no one-size-fits-all approach, as urban areas face higher pollution exposure while certain rural counties show elevated health risks due to demographic factors. Indexes such as the Air Health Vulnerability Index (AHVI) and the risk score can provide additional insights, although more sophisticated statistical methods may be needed. Overall, air quality analysis can inform the works of many, including policymakers in protecting socially vulnerable populations during air quality events, and more generally the well-being of the community.

Limitations

We would also like to address some limitations inherent in our analysis.

The AQI dataset did not include all U.S. counties, which limits the completeness of county-level comparisons and may bias results toward regions with more extensive air quality monitoring infrastructure. Also, the analysis relies on a single year of AQI data (2025), which may not account for year-to-year variability driven by events such as wildfires, extreme weather, or policy changes.